

A. Assignment Information

Particular	Details
Department:	Department of Computer Science and Engineering
Programme:	B.Tech – Computer Science and Engineering (CSE) – Artificial Intelligence
Course Code & Course Name	DSA0401-Fundamentals of Data Science
Academic Year / Batch	3 rd year / 2024-28
Faculty Name	Dr. T. P. Anithaashri
Assignment Title	Urban Parking Availability Prediction Using Statistical and Machine Learning Techniques
Date of Issue	31/08/2026
Date of Submission	31/08/2026
Maximum Marks	100
Course Outcome(s) – CO	CO4 – Statistical Inference; CO5 – Supervised and Unsupervised Machine Learning
Bloom's Taxonomy Level	L3 – Apply, L4 – Analyze, L5 – Evaluate
SDG Mapping (SDG 1-17)	SDG 11 – Sustainable Cities and Communities
Industry / Societal Relevance	Smart-city parking management, urban mobility, sustainable transportation, and data-driven infrastructure planning

B. Assignment Problem / Challenge

Urban parking facilities generate and maintain a large amount of information related to their location, structure, parking system, parking restrictions, number of parking decks, gantry height, and basement availability. However, raw parking data cannot be directly used for meaningful analysis without proper preprocessing and analytical methods. The challenge of this project is to develop a data-driven urban parking analysis and predictive analytics system using the HDBCarparkInformation.csv dataset. The system should clean and preprocess the dataset, identify important patterns through exploratory data analysis, apply statistical inference, and use supervised and unsupervised machine learning techniques to analyze parking characteristics.

The project specifically applies point estimation, confidence intervals, and hypothesis testing to understand the statistical properties of the dataset. It also applies Linear Regression, Logistic Regression, kNN, and Decision Tree/CART for predictive and classification tasks. K-Means clustering is used to group similar parking facilities based on their characteristics. Ten model iterations are compared using suitable evaluation metrics to identify an effective classification approach. The solution should provide meaningful visualizations, statistical results, model performance measures, and cluster analysis. The final results can help demonstrate how data science and predictive analytics can support urban parking management, smart-city planning, and sustainable transportation decisions.

Main Challenges

1. Cleaning and preparing real-world parking data.
2. Handling missing and duplicate records.
3. Converting categorical parking information into machine-learning features.
4. Applying statistical inference correctly.
5. Building and comparing multiple machine learning models.
6. Performing 10 model iterations with different algorithm parameters.
7. Selecting a suitable model based on quantitative evaluation.
8. Identifying similar parking facilities using K-Means clustering.
9. Interpreting results for practical urban parking applications.
10. Clearly identifying the limitation that the supplied dataset does not contain real-time parking occupancy or vacant-space information.

C. Problem Statement

Urban parking management requires effective analysis of parking facilities and their characteristics. The available parking dataset contains information such as location coordinates, car park type, parking system, parking restrictions, number of parking decks, gantry height, and basement availability. However, these raw records need to be properly cleaned, transformed, and analyzed before useful patterns can be identified. The problem addressed in this project is to develop a data science and predictive analytics framework for urban parking analysis using the HDBCarparkInformation.csv dataset. The system aims to apply statistical inference techniques to understand parking characteristics and use machine learning techniques to perform prediction, classification, and clustering.

The project uses point estimation, confidence intervals, and hypothesis testing to analyze the statistical properties of parking data. It applies Linear Regression for numerical prediction and Logistic Regression, kNN, and Decision Tree/CART for classification. K-Means clustering is used to identify groups of car parks having similar characteristics. The performance of the classification models is evaluated through 10 iterations with different model parameters. Metrics such as accuracy, precision, recall, and F1-score are used to compare the models, while regression is evaluated using MAE, RMSE, and R^2 . For clustering, the silhouette score is used to select a suitable number of clusters. The final system provides a structured approach for understanding urban parking data and demonstrates how statistical and machine learning techniques can support data-driven parking management and smart-city planning.

Problem Definition

To develop a data-driven system that analyzes urban parking facility characteristics are using statistical inference, regression, classification, and K-Means clustering techniques, and evaluates multiple machine learning models to identify useful patterns and predictive insights for improved urban parking management.

D. Requirements and Constraints

Requirements

The proposed urban parking data analysis system requires a structured dataset containing information about parking facilities. The system uses the HDBCarparkInformation.csv

dataset and performs data preprocessing, statistical analysis, machine learning, and clustering. The assignment format specifies that relevant functional, technical, resource, sustainability, and other constraints should be identified according to the particular problem.

The main requirements of the project are:

1. **Dataset Requirement:** The system shall use the provided HDBCarParkInformation.csv dataset as the primary source of data.
2. **Data Preprocessing:** The system shall identify and handle duplicate records, missing values, incorrect data types, and invalid numerical values.
3. **Feature Engineering:** Relevant categorical parking attributes shall be converted into numerical or binary features suitable for machine learning algorithms.
4. **Exploratory Data Analysis:** The system shall generate descriptive statistics, distributions, correlation analysis, and appropriate visualizations.
5. **Statistical Inference:** The system shall calculate a point estimate, confidence interval, and perform hypothesis testing using a p-value.
6. **Regression Requirement:** Linear Regression shall be implemented for predicting a numerical parking-related attribute.
7. **Classification Requirement:** Logistic Regression, kNN, and Decision Tree/CART shall be implemented for classification.
8. **Ten Iterations:** Ten supervised machine learning configurations shall be trained and compared using appropriate evaluation metrics.
9. **Clustering Requirement:** K-Means clustering shall be implemented to identify groups of similar parking facilities.
10. **Model Evaluation:** Classification models shall be evaluated using accuracy, precision, recall, and F1-score. Regression shall use MAE, RMSE, and R^2 , while clustering shall use silhouette score.
11. **Result Storage:** Processed data and analytical results shall be saved as CSV files for further use and documentation.

Constraints

1. **Dataset Constraint:** The supplied dataset contains static parking-facility information and does not contain real-time parking occupancy or vacant-space values.

2. **Prediction Constraint:** The current implementation therefore cannot make a genuine real-time prediction of the number of available parking spaces.
3. **Data Quality Constraint:** Missing or inconsistent values may affect the accuracy of statistical analysis and machine learning models.
4. **Feature Constraint:** Only variables available in the supplied dataset can be used unless an additional dataset is introduced.
5. **Computational Constraint:** The models should be computationally efficient enough to execute on a standard computer or cloud notebook environment.
6. **Data Leakage Constraint:** The target variable must not be directly included among the input features used to train its prediction model.
7. **Evaluation Constraint:** Model performance must be based on actual execution results rather than manually assumed or fabricated values.
8. **Sustainability Constraint:** The analysis should support the broader goal of efficient urban mobility and sustainable parking management.
9. **Interpretability Constraint:** Model results should be presented in a manner that can be understood and interpreted for practical parking-management applications.
10. **Future Data Constraint:** Direct parking availability forecasting would require additional historical, time-stamped occupancy or availability data.

E. Student Work

1. Problem Understanding and Formulation

Urban parking facilities have different characteristics such as location, parking type, parking system, number of decks, gantry height, and basement availability. Analyzing these factors using data science can help understand parking patterns and support better urban parking management.

This project uses the HDBCarparkInformation.csv dataset to perform data preprocessing, statistical analysis, prediction, classification, and clustering. Statistical techniques such as point estimation, confidence intervals, and hypothesis testing are applied, followed by Linear Regression, Logistic Regression, kNN, Decision Tree/CART, and K-Means clustering.

Expected Outcomes

1. Clean and preprocess the parking dataset.

2. Analyze parking characteristics using graphs and statistics.
3. Calculate confidence intervals and perform hypothesis testing.
4. Apply Linear Regression for numerical prediction.
5. Compare 10 machine learning iterations.
6. Apply Logistic Regression, kNN, and Decision Tree/CART.
7. Group similar car parks using K-Means.
8. Evaluate models using suitable performance metrics.

Assumptions

1. Each car park number represents a unique facility.
2. Missing numerical values can be handled using median values.
3. The two most common parking-system categories can be used for classification.
4. Standardized features are suitable for kNN and K-Means.

Dataset Limitation

The dataset contains static parking information and does not provide real-time occupancy or available-space values. Therefore, actual real-time parking availability prediction would require additional historical occupancy data. This follows the assignment format's requirement to explain the problem, expected outcomes, available data, assumptions, and constraints.

2. Application of Course Knowledge

The project applies the concepts of **statistical inference and machine learning** to analyze urban parking data.

1. **Point Estimation:** Calculates the mean number of parking decks.
2. **Confidence Interval:** Estimates the 95% confidence range for the mean number of decks.
3. **Hypothesis Testing:** Uses a one-sample t-test and p-value to test the selected hypothesis.
4. **Linear Regression:** Predicts gantry height using parking-related features.
5. **Logistic Regression:** Classifies the parking-system category.
6. **kNN:** Classifies parking systems based on similar observations.
7. **Decision Tree/CART:** Performs classification using decision rules and Gini criterion.

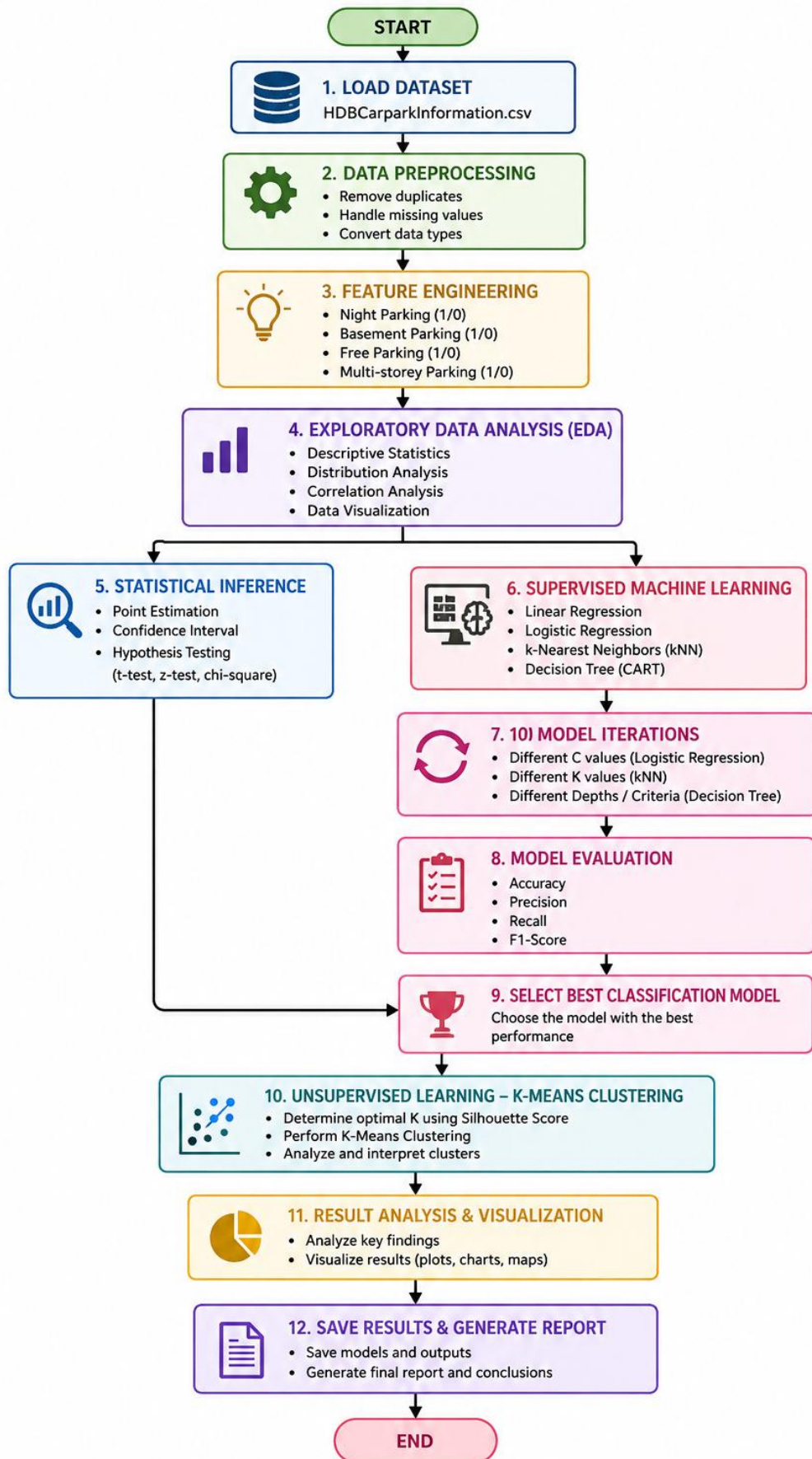
8. **K-Means:** Groups car parks with similar characteristics.
9. **Evaluation Metrics:** Uses MAE, RMSE, R^2 , accuracy, precision, recall, F1-score, and silhouette score.

3. Solution / Design / Methodology

The proposed system follows a step-by-step data science workflow. The parking dataset is first loaded and cleaned by removing duplicates, handling missing values, and converting data into suitable formats. Relevant categorical attributes are converted into numerical features. The cleaned data is then analyzed using exploratory data analysis, including descriptive statistics, graphs, and correlation analysis. Statistical inference is performed using point estimation, 95% confidence interval, and hypothesis testing. For predictive analysis, Linear Regression is applied to a numerical parking characteristic. For classification, Logistic Regression, kNN, and Decision Tree/CART are implemented. Ten different model configurations are trained and compared using accuracy, precision, recall, and F1-score. Finally, K-Means clustering is applied to group similar car parks. Different values of K are tested using the silhouette score, and the best clustering configuration is selected. The results are visualized and saved as CSV files for further analysis.

Methodology Flow

Dataset → Data Cleaning → Feature Engineering → EDA → Statistical Inference → Regression → Classification → 10 Model Iterations → K-Means Clustering → Evaluation → Result Analysis



4. Use of Modern Tools

The project uses Python-based data science tools for data preprocessing, statistical analysis, machine learning, and visualization.

1. **Python:** Main programming language.
2. **Pandas:** Data loading, cleaning, and data manipulation.
3. **NumPy:** Numerical calculations and mathematical operations.
4. **SciPy:** Confidence intervals and hypothesis testing.
5. **Scikit-learn:** Regression, classification, clustering, preprocessing, and model evaluation.
6. **Matplotlib:** Creating analytical graphs.
7. **Seaborn:** Creating statistical visualizations.
8. **Jupyter Notebook / Google Colab / VS Code:** Running and testing the project code.

5. Results and Validation

The developed system produces statistical results, visualizations, prediction results, classification performance, and clustering results. The models are evaluated using suitable performance metrics.

1. **Statistical Analysis:** Mean, 95% confidence interval, t-statistic, and p-value are calculated.
2. **Linear Regression:** Evaluated using **MAE, RMSE, and R^2** .
3. **Classification:** Logistic Regression, kNN, and Decision Tree/CART are compared using **accuracy, precision, recall, and F1-score**.
4. **10 Iterations:** Ten different model configurations are executed and their performance is compared.
5. **K-Means:** Different values of K are tested using the silhouette score.
6. **Visualization:** Graphs, confusion matrix, feature importance, and cluster plots are generated.

The best classification model is selected based on its performance, particularly the F1-score. The actual numerical values will be obtained after executing the program with the dataset.

This ensures that the reported results are based on actual model output rather than assumed values.

6. Analysis and Engineering Decision

The results from the statistical and machine learning analysis are compared to identify useful patterns in the parking dataset. The 10 classification iterations are evaluated using accuracy, precision, recall, and F1-score. The model with the best overall performance is considered the preferred classification model. Linear Regression results are analyzed using MAE, RMSE, and R^2 to determine the quality of numerical prediction. For K-Means, the silhouette score is used to select a suitable number of clusters and identify groups of similar parking facilities. The analysis helps determine which features and algorithms provide better results. Decision Tree feature importance also helps identify the parking characteristics that contribute most to classification. The final decision is based on the actual performance values obtained from the executed program.

The main advantages of the proposed approach are its simple implementation, multiple model comparison, measurable evaluation, and combination of supervised and unsupervised learning. Its major limitation is that the supplied dataset does not contain real-time occupancy information, so direct vacant-space prediction cannot be validated. This follows the assignment requirement to compare alternatives, identify advantages and limitations, discuss trade-offs, and justify the final solution using quantitative or qualitative evidence.

7. Broader Considerations

The proposed urban parking analysis system can contribute to sustainable urban mobility and smart-city development by providing data-driven insights into parking facilities.

1. **Sustainability:** Better parking analysis can support efficient use of existing parking infrastructure.
2. **Environment:** Efficient parking management can potentially reduce unnecessary vehicle movement and fuel consumption.
3. **Society:** Improved parking planning can provide a better experience for vehicle users and urban planners.

4. **Economic Impact:** Data-based decisions can help optimize the use and planning of parking infrastructure.
5. **Professional Responsibility:** Model results should be interpreted carefully, and limitations of the available dataset must be clearly communicated.
6. **Ethics:** The system should use reliable data and avoid making unsupported claims about real-time parking availability.

8. Conclusion

The project successfully demonstrates the application of data science and predictive analytics techniques to urban parking data. The system performs data preprocessing, exploratory analysis, statistical inference, regression, classification, and clustering using the available parking dataset. Statistical methods such as point estimation, confidence intervals, and hypothesis testing provide quantitative insights into parking characteristics. Linear Regression, Logistic Regression, kNN, and Decision Tree/CART demonstrate supervised learning, while K-Means identifies groups of similar parking facilities. The **10 model iterations** provide a systematic comparison of different machine learning configurations. The best-performing model can be selected based on the obtained evaluation metrics. Overall, the project provides a structured approach for analyzing urban parking facilities and demonstrates the usefulness of data-driven methods for smart-city planning. The major limitation is that the supplied dataset does not contain historical or real-time parking occupancy/availability data. Therefore, direct vacant-space prediction cannot be established using this dataset alone. In the future, historical occupancy data can be incorporated to develop a true parking availability forecasting system.

9. Student Reflection

This assignment helped me understand how data science techniques can be applied to a real-world urban parking problem. I learned how raw data needs to be cleaned and transformed before applying statistical and machine learning methods. I gained practical experience in using statistical inference, Linear Regression, Logistic Regression, kNN, Decision Tree/CART, and K-Means clustering. Comparing 10 different model iterations also helped me understand how model parameters can affect performance and why evaluation metrics are important when selecting a suitable model. If additional time and resources were available, I would include historical parking occupancy, available-space, traffic, weather, and

time-based information. This would allow the system to move from static parking analysis toward a more accurate real-time parking availability prediction system.

10. References

- [1] S. X. Xiao, Z. Peng, Y. Lin, Z. Jin, W. Shao, R. Chen, N. Cheng, and G. Mao, "Parking prediction in smart cities: A survey," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 10, 2023, doi: 10.1109/TITS.2023.3279024.
- [2] S. S. Channamallu, S. Kermanshachi, J. M. Rosenberger, and A. Pamidimukkala, "Enhancing urban parking efficiency through machine learning model integration," *IEEE Access*, vol. 12, pp. 81338–81347, 2024, doi: 10.1109/ACCESS.2024.3411391.
- [3] S. Bilotta, L. A. Ipsaro Palesi, and P. Nesi, "Predicting free parking slots via deep learning in short-mid-terms explaining temporal impact of features," *IEEE Access*, vol. 11, 2023, doi: 10.1109/ACCESS.2023.3314660.
- [4] S. Gong, J. Qin, H. Xu, R. Cao, Y. Liu, C. Jing, Y. Hao, and Y. Yang, "Spatio-temporal parking occupancy forecasting integrating parking sensing records and street-level images," *International Journal of Applied Earth Observation and Geoinformation*, vol. 118, p. 103290, 2023, doi: 10.1016/j.jag.2023.103290.
- [5] J. Fan, Q. Hu, Y. Xu, and Z. Tang, "Predicting vacant parking space availability: A long short-term memory approach," *IEEE Intelligent Transportation Systems Magazine*, vol. 14, no. 2, pp. 129–143, 2022, doi: 10.1109/MITS.2020.3014131.
- [6] S. Saharan, N. Kumar, and S. Bawa, "DyPARK: A dynamic pricing and allocation scheme for smart on-street parking system," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 4, pp. 4217–4234, 2023, doi: 10.1109/TITS.2022.3230851.
- [7] S. Channamallu *et al.*, "Examining parking occupancy prediction models: A comparative analysis," *Transportation Research Procedia*, 2023.
- [8] C. Ma, "A review of research on urban parking prediction," *Journal of Traffic and Transportation Engineering (English Edition)*, 2024, doi: 10.1016/j.jtte.2023.11.004.

[9] M. S. et al., “Predicting open parking space using deep learning and support vector regression,” in *Proc. 2023 7th Int. Conf. Intelligent Systems, Metaheuristics and Swarm Intelligence*, 2023, pp. 20–27, doi: 10.1145/3596947.3596961.

[10] “Deep learning for on-street parking violation prediction,” in *Proc. 2022 IEEE 14th Image, Video, and Multidimensional Signal Processing Workshop (IVMSP)*, 2022, doi: 10.1109/IVMSP54334.2022.9816222.

F. APPENDICES

Sample Python Code:

```
# Urban Parking Analysis
```

```
import pandas as pd
from scipy import stats
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.cluster import KMeans
from sklearn.metrics import accuracy_score, f1_score, r2_score

# Load dataset
df = pd.read_csv("HDBCarparkInformation.csv")

# Data cleaning
df = df.drop_duplicates()
df["car_park_decks"] = pd.to_numeric(
    df["car_park_decks"], errors="coerce"
)
df["car_park_decks"] = df["car_park_decks"].fillna(
```

```

    df["car_park_decks"].median()
)

# Statistical inference
mean_decks = df["car_park_decks"].mean()
t_stat, p_value = stats.ttest_1samp(
    df["car_park_decks"], 2
)
print("Mean:", mean_decks)
print("p-value:", p_value)

# Features
X = df[["x_coord", "y_coord", "car_park_decks"]]
y = df["car_park_decks"]

# Linear Regression
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = LinearRegression()
model.fit(X_train, y_train)
pred = model.predict(X_test)
print("Regression R2:",
      r2_score(y_test, pred))

# Classification example
df["target"] = (
    df["type_of_parking_system"] ==
    df["type_of_parking_system"].value_counts().index[0]
).astype(int)
X = df[["x_coord", "y_coord", "car_park_decks"]]

```

```

y = df["target"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2,
    random_state=42,
    stratify=y
)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# 10 model iterations
models = [
    LogisticRegression(C=1),
    LogisticRegression(C=0.1),
    LogisticRegression(C=10),
    KNeighborsClassifier(n_neighbors=3),
    KNeighborsClassifier(n_neighbors=5),
    KNeighborsClassifier(n_neighbors=7),
    DecisionTreeClassifier(max_depth=3),
    DecisionTreeClassifier(max_depth=5),
    DecisionTreeClassifier(max_depth=8),
    DecisionTreeClassifier(criterion="entropy")
]
for i, model in enumerate(models, 1):
    model.fit(X_train, y_train)
    prediction = model.predict(X_test)
    print(
        "Iteration", i,
        "Accuracy:",
        round(accuracy_score(y_test, prediction), 3),
        "F1:",

```

```

        round(f1_score(y_test, prediction), 3)
    )

# K-Means clustering
kmeans = KMeans(
    n_clusters=3,
    max_iter=10,
    random_state=42,
    n_init=10
)
clusters = kmeans.fit_predict(
    scaler.fit_transform(X)
)
df["cluster"] = clusters
print("\nK-Means clustering completed.")

```

Output:

